

# Deep Learning: Homework 4

June 8, 2021

1. Your task is to train a classifier on **cifar100** dataset. You can see information about data by following this [link](#). You should do the following steps in your homework:
  - a) Use vgg or vgg like architecture as your model without batch normalizations.
  - b) Use the model of previous point with batch normalizations in all convolutional layers putting them before activation functions.
  - c) Use the model of the point a) with batch normalizations in all convolutional layers putting them after activation functions.
  - d) For all above points calculate top1 and top5 accuracies on the test dataset and compare them.
2. In this point you should write your own class for dense layer inherited from **tf.Module**. An instance of that class should be callable i.e. you should implement a `__call__` method. Arguments of the method `__init__` should be *input\_size*, *output\_size* and *activation*. In this method you should implement the initialization of the kernel (W) and bias of your layer (Glorot initialization for kernel and initialization with zeros for bias).